

Kunal Ramdas Jadhav

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Professional Summary

Aspiring AI Engineer pursuing B.Tech in Robotics and Artificial Intelligence at COEP Technological University, with a strong foundation in computer vision, machine learning, and intelligent systems development. Demonstrated ability to design and deploy end-to-end AI solutions — from real-time object detection pipelines using YOLOv8 to LLM-integrated copilots and predictive analytics platforms. Experienced in building production-grade systems with FastAPI, Docker, and AWS, with a particular focus on real-time video analytics, geo-spatial intelligence, and AI-driven automation. Recognized across multiple national and international hackathons for delivering innovative AI solutions under competitive conditions.

Career Objective: To contribute to Trinetra Aerospace Technologies as an AI Engineer, leveraging expertise in computer vision, real-time video feed processing, and location-aware analytics to build intelligent systems that advance aerospace and defense-grade operational capabilities.

Education

COEP Technological University <i>B.Tech in Robotics and Artificial Intelligence</i>	CGPA: 7.90 <i>2025 – 2028</i>
Cusrow Wadia Institute of Technology, Pune <i>Diploma in Computer Engineering</i>	95.53% <i>2025</i>
S.J.M.V. Jadhavwadi Bori <i>Secondary School Certificate (SSC)</i>	90.80% <i>2022</i>

Technical Skills

AI / ML	TensorFlow, PyTorch, Scikit-Learn, XGBoost, Random Forest, Prophet, YOLO (v5/v8), OpenCV
LLM & AI Systems	LLM integration, AI Copilot design, RAG pipelines, natural language interfaces for AI
Computer Vision	Object detection & tracking, real-time video analytics, crowd analysis, geo-tagged video processing
Backend	FastAPI, Python, REST APIs, WebSockets, Firebase, PostgreSQL, Redis
Frontend	React, TypeScript, Next.js, Tailwind CSS, HTML
MLOps & DevOps	Docker, AWS, Git, GitHub, CI/CD workflows, microservices architecture
Languages	Python, C++, C, Java, SQL
Tools	Google OR-Tools, Google Maps API, Android Studio, Jupyter Notebook

Projects

AI Guardian – Real-Time Crowd Monitoring & Stampede Prediction System 2025
Computer Vision · YOLO · Anomaly Detection · Real-Time Video Analytics

- Built an AI system analyzing live CCTV/video feeds for crowd density estimation, anomaly detection, and early stampede risk prediction — directly applicable to geo-tagged surveillance feed analysis.
- Developed a real-time computer vision pipeline using **YOLOv8** for multi-person detection, tracking, and spatial clustering across video frames.
- Engineered spatial-temporal features capturing crowd velocity vectors, density gradients, and congestion patterns to identify pre-stampede behavioral signatures.
- Implemented heatmap overlays, real-time alert triggers, and a hybrid rule-based + ML risk scoring model for proactive incident response.
- github.com/mahek-s11/ai-stampede-early-warning-system

Network Traffic Anomaly Detection – ML-Based Intrusion Detection System June 2025
Random Forest · XGBoost · Ensemble Learning · Real-Time Classification

- Designed a hybrid anomaly detection system combining statistical profiling with an ensemble of Random Forest and XGBoost models for real-time classification of network threats.
- Engineered network-flow features (packet rate, byte rate, flow duration, inter-arrival time, source-destination imbalance) for high-signal traffic characterization.
- Achieved high precision, recall, and ROC-AUC scores; validated system for low-latency real-time monitoring suitability.
- github.com/kunalj2558/Detecting-Unbalanced-Network-Traffic

Early Plant Stress Detection using Bio-ECG Signals & Digital Twin 2025
Signal Processing · Embedded Sensors · Digital Twin · Predictive Analytics

- Built an intelligent monitoring system capturing real-time micro-voltage signals from plant tissues via embedded sensors, with ML models identifying stress conditions before visible symptoms.
- Constructed a Digital Twin of the plant to simulate growth trajectories and forecast future stress scenarios, enabling

proactive precision agriculture intervention.

AI Project Experience

SmartDepot AI – Transport Operations Intelligence Platform

Jan 2026 – Present

FastAPI · PostgreSQL · XGBoost · Prophet · Google OR-Tools · React · Docker · AWS

- Designed and developed an AI-powered platform for MSRTC integrating crew scheduling, fleet allocation, and predictive maintenance - deployed across multiple depots using Docker microservices on AWS.
- Built predictive ML modules (XGBoost, Random Forest, Prophet) for crew shortage forecasting, demand prediction, and vehicle maintenance risk scoring.
- Engineered an **AI Copilot** enabling natural language queries over scheduling, maintenance, and workforce data - demonstrating end-to-end LLM integration in an operations context.
- Developed executive analytics dashboards for KPI monitoring: fleet utilization, route performance, workforce availability, and maintenance risk indicators.

ChargeX AI – Intelligent EV Charging Interoperability Platform

May 2026 – Present

FastAPI · PostgreSQL · Redis · WebSockets · XGBoost · Prophet · Next.js · Docker · AWS

- Architected a scalable microservices backend with real-time WebSocket-based charger synchronization supporting high-frequency operational updates across distributed charging networks.
- Developed ML models for charger availability prediction, queue-time estimation, demand forecasting, and predictive maintenance - improving station utilization and operational planning.
- Built an intelligent EV route optimization engine factoring battery SoC, range, real-time charger availability, traffic conditions, and charging costs using Google Maps API.
- Engineered an **AI-powered EV Copilot** supporting natural-language queries for route planning, infrastructure monitoring, and operational decision-making.

Internship & Experience

Research Intern – Computer Vision & Autonomous Systems

May 2026 – Present

COEP Technological University

- Researching and developing an AI-based **Bike ADAS** platform for intelligent two-wheeler rider safety applications.
- Implementing real-time perception pipelines using camera and radar sensor fusion for obstacle detection, depth estimation, and collision prediction.
- Building simulation environments in **CARLA** to evaluate ADAS functionalities - lane departure, emergency braking, and hazard detection - across diverse traffic scenarios.
- Exploring advanced computer vision, sensor fusion, and ML techniques to architect next-generation autonomous safety systems for two-wheelers.

Android Development Intern

May 2024 – July 2024

Pro-Azure - Mentored by Ganesh Khaplae

- Developed Android applications in Java with XML-based UI; integrated REST APIs and Firebase for backend data management.
- Practiced debugging, performance profiling, and collaborative development using Git/GitHub.

Achievements & Recognition

- **Runner-Up (2nd Place)** – AI for Bharat 2.0 Hackathon (Govt. of Karnataka, PAN IIT & AWS); built AI solution for public-sector challenges, competing against teams nationwide.
- **Winner** – HackArena 2026 (Ignite Room & Polaris School of Technology, Bengaluru).
- **Top 8 Globally** – Cavesta International Hackathon Finalist; hackathon ran simultaneously across 4 countries.
- **National Finalist** – RECKON 7.0, Rajasthan University, Jodhpur; represented COEP Technological University.
- **State Representative** – Aavishkar 2026 Research Convention; presented research at state-level inter-university competition.

Leadership & Extracurriculars

- **Class Representative** - Computer Engineering Dept., Second Year to Final Year, COEP Technological University.
- **Automation Coordinator** – Aerial Robotics Study Circle (ARSC); worked on autonomous drone navigation and mission execution systems.
- **T&P Student Coordinator** – Assisted in coordinating recruitment drives and industry-student interactions at COEP.
- **HackerRank Campus Crew** – Promoted competitive programming and coding culture at COEP Technological University.
- **Impression Media Coordinator** – Managed digital outreach and media partnerships for college technical festivals.